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SERVICE NOTE

13-MAR-2013

Class A

SN DOC1236890, Rev 1

New HEC Cabinet G6001EN

EFFECTIVITY:

ALL DV Heat Exchanger Cabinets (HEC)

PROBLEM:

There have been several customer requests to widen the allowable input water temperature range for the GE Heat Exchanger Cabinet (HEC) from 7C – 10C to 6C - 12C.

This service note also alerts the field that the Pump Shield was removed in new HEC cabinets as it is no longer needed.

SOLUTION:

MR450w and MR750w sites equipped with new HEC catalog number G6001EN combined with "DV23.1_V01 with ServicePack01" or later software, permit use of new 6-12 C inlet water requirement. New Guided Install selection entry under Hardware Configuration page permits FE to select G6001EN HEC at sites where it is present. There will be no change at MR450, MR750 & MR950 sites equipped DV23.1 software with new G6001EN HEC as these will continue to adhere to existing 7-10C inlet water temperature requirement.

The HEC G6000EN will phase out the end of Q1 2013 and the new HEC G6001EN will phase in Q2 2013. **THERE IS NOT AN OPTION TO UPGRADE A SITE WITH A G6000EN TO A G6001EN.**

Facility Water Inlet Temperature		
Product	HEC G6000EN	HEC -NEW G6001EN**
MR750w	7-10 C	6-12 C
MR450W		
MR750		7-10 C
MR450		
MR950		
**DV23.1_V01 with SP01 or later software is needed for the NEW G6001EN cabinet		

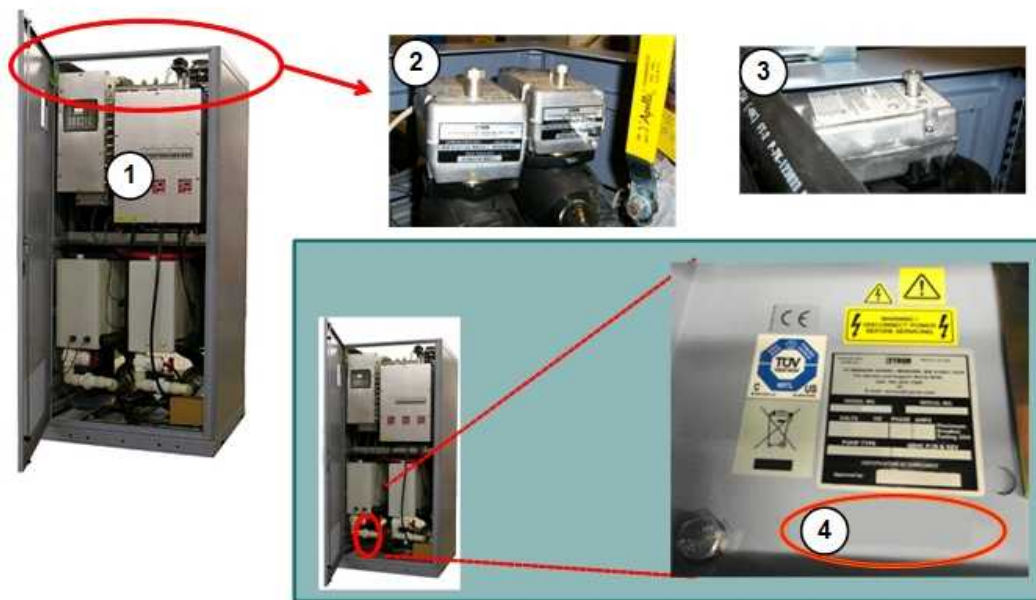
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HARDWARE CHANGES:

The hardware changes below have been made to the HEC G6001EN to allow for the expanded temperature range. Illustration 1 and 2 below describe how to identify HEC type.

- Rev 3 of GC / PE mixing valves 5156318. This is a new valve; reference illustration below.
- Rev 3 of Chilled Air Blower mixing valves 5156320. This is a new valve; reference the illustration below.
- Rev 5 of Signal box 5269691. Internal component changes have been made.
- Rev 3 of Fuse Kit 5266827. Change was to add fuse T2AL250V as this is needed with the new mixing valve 5156318 rev 3.
- Rev 3 of Cryo Supply Header 5376555. Change was that the flow regulator in Cryo Supply Header changed to 2.25gpm.

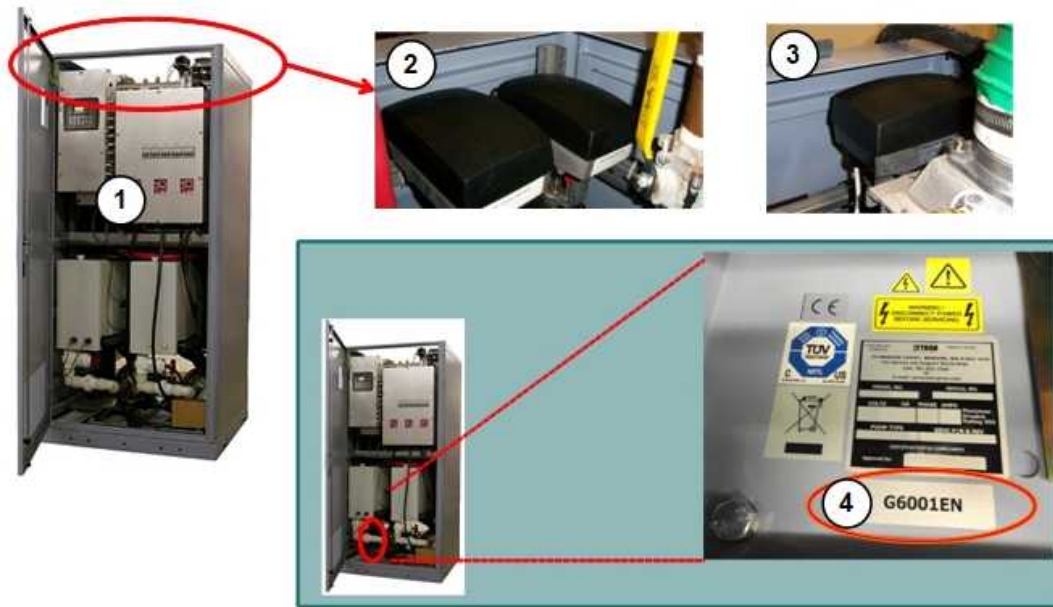
Illustration 1: HEC Type G6000EN



1	HEC catalog number G6000EN	3	Chilled air mixing valve
2	GC/PE mixing valve	4	No G-catalog label

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Illustration 2: HEC Type G6001EN



1	HEC catalog number G6001EN	3	Chilled air mixing valve
2	GC/PE mixing valve	4	Catalog label G6001EN

FRU Compatibility

All earlier versions of affected FRU parts have been purged from stock and replaced with the latest revision. See below table for part compatibility.

Part Description / ID	G6000EN (Existing)	G6001EN (New)
GC / PE mixing valves 5156318 Rev2	YES	NO*
GC / PE mixing valves 5156318 Rev3	YES	YES
Chilled Air Blower mixing valves 5156320 Rev2	YES	NO*
Chilled Air Blower mixing valves 5156320 Rev3	YES	YES
Signal box 5269691 Rev4	YES	NO*
Signal box 5269691 Rev5	YES	YES
Fuse Kit 5266827 Rev2	YES	NO*
Fuse Kit 5266827 Rev3	YES	YES
Cryo Supply Header 5376555 Rev2	YES	NO*
Cryo Supply Header 5376555 Rev2	YES	YES
*Earlier revisions of these parts have been removed from GPRS stock and replaced with the most recent revision		

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Revised Service Procedures

Appendix A contains the revised valve and signal box replacement procedures. The procedures listed below had minor edits to describe the change detailed above. All will be included in the next release of the PIM / Service Methods DVD's.

- MR750W Pre-Installation Manual: 5670003TPH
- MR450W Pre-Installation Manual: 5670001TPH
- 820-0190 Heat Exchanger Cabinet Installation and Service vendor manual
- HEC Troubleshooting – Discovery & Optima service methods
- Heat Exchanger Theory – Discovery & Optima service methods
- HEC Facility Tee with QD Replacement (removal of pump shield) – Discovery & Optima service methods
- Heat Exchanger Cabinet Setup - Discovery & Optima service methods
- MR450w and MR750w FRU manual - Discovery & Optima service methods

Rebecca David

Lead Service Integrator
Service/Systems Engineering
GE Healthcare – Americas

Yoshiaki Tsujii

Service Modality Engineer
Service Operations
GE Healthcare - Asia

Carlos Lenero

Service Quality / Productivity Leader
Service Operations EMEA
GE Healthcare – Europe

Bill Bonneau

Cooling Architect
MR Premeium Engineering
GE Healthcare – Americas

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APPENDIX A:

- 820-0339 – Installation Instructions GC + PE Mixing Valves (CLS VSID 1067614.pdf)
- 820-0340 – Installation Instructions Chilled Air Blower Mixing Valve (CLS VSID 1067615.pdf)
- 820-0234 - Signal Box (CLS VSID 1067618.pdf)



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820-0339 – Replacement Instructions GC and PE Mixing Valves PN 205-0121-30 (GE P/N 5156318)

Purpose: The purpose of this document is to provide instruction on how to remove and replace the GC and PE mixing valves. Note that there are 2 different types of valves (refer to illustrations within this document to determine mixing valve type). The mixing valve will be supplied with a mounting bracket and hoses making installation and removal in all cabinets possible. The GC and PE valve assemblies in a new cabinet are identical in form, only the hose lengths are different. This kit has been designed with hose lengths to replace either a GC or PE valve.

Scope: This instruction applies to all DVMR Heat Exchanger Cabinets (HEC).

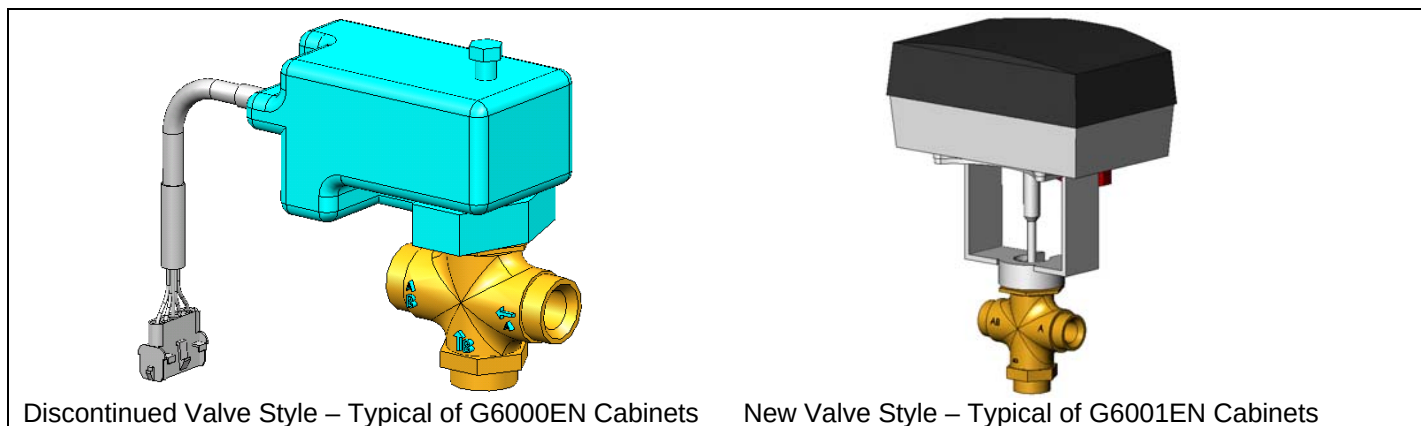
Tools Required:

- 1-1/4" open end or adjustable wrench
- Adjustable wrench (opening up to 1-1/8" for backing stationary fittings)
- 7/16" deep welled socket or wrench
- Absorbent towels
- Wet/Dry Vacuum or Bucket for draining

People Required: 2

Time Required: 2 hours per valve

Refer to the illustrations below to determine mixing valve type.



Discontinued Valve Style – Typical of G6000EN Cabinets

New Valve Style – Typical of G6001EN Cabinets

Illustration 1 – Mixing Valve Styles

Removal Procedure:

1. Perform shutdown and lockout of system in accordance with the GE "LOTO for Heat Exchanger Cabinet (HEC)" procedure.
2. Open door to heat exchanger cabinet. If necessary, the door can be removed from its hinges by disconnecting the ground strap (The ground strap nut can either be removed using a 10mm or adjustable wrench) and lifting up on the door.

3. Close Facility Supply and Return Ball valves located at the top of the HEC to isolate HEC from facility water.
4. Locate either GC or PE mixing valve (Refer to Illustration 2).

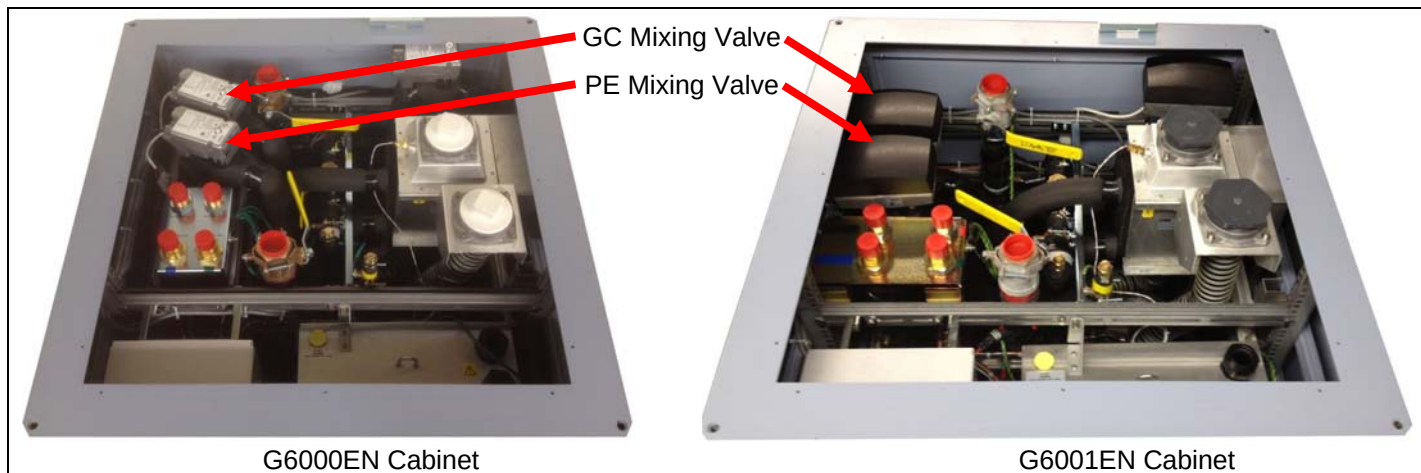


Illustration 2 – GC and PE Mixing Valve Locations in G6000EN and G6001EN Cabinets

5. Drain Facility side plumbing per GE draining procedures.

Mixing Valve Assembly Kit

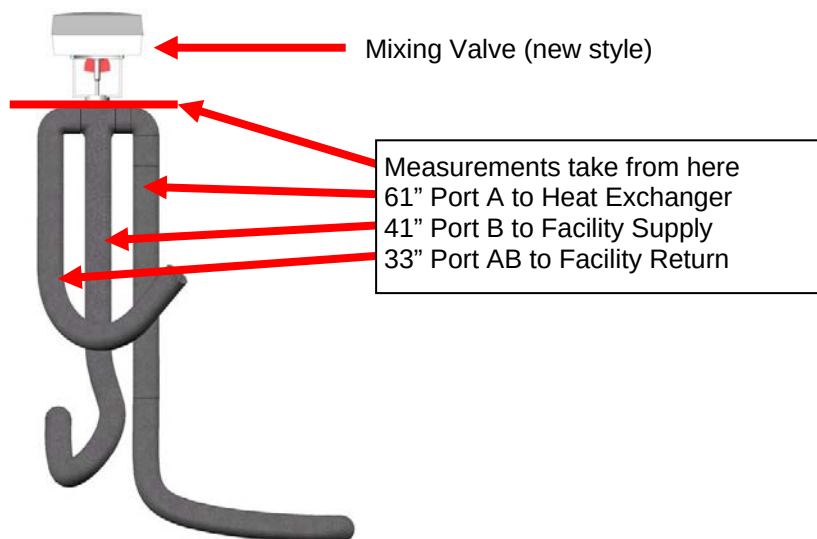


Illustration 3 - Connections for Either the GC or PE Mixing Valve Kit



CAUTION When removing items weighing greater than 35lbs, the hoist system must be utilized. Failure to use the hoist system may result in damage to the equipment or personal injury.

6. It is recommended to move/remove either the Signal Box or the Electrical Box to gain access to connection points on Facility Return header. In order to slide the Electrical Box to the side, disconnect the cables between the Electrical Box and the Signal Box. Refer to Lytron Procedures #820-0233 (Electrical Box) and #820-0234 (Signal Box) for details on removing these parts. It is also recommended to remove the blower to get better access to the hoses per GE main Blower replacement procedure under Heat Exchanger Cabinet Setup.

7. The Photos below detail the locations for connecting ports A, AB, and B to the system. Refer to Illustration 3 above for locations of ports detailed in following photos. NOTE: Both Signal Box and Electrical Box have been removed in the photos below for ease of locating connections.

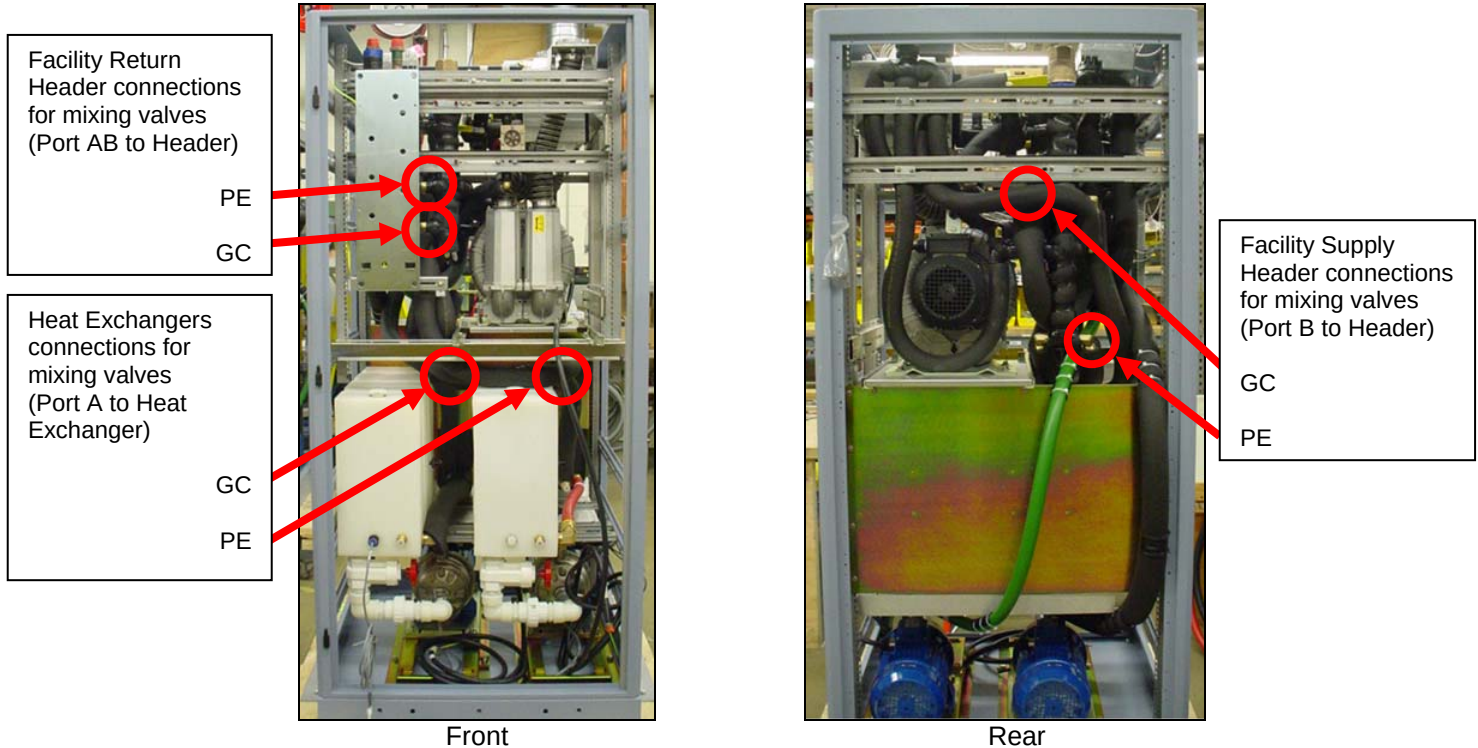


Illustration 4 - Detail of Cabinet with Signal Box and Electrical Box removed (Front and Rear Views)

8. Remove 5 pin connector for appropriate mixing valve (GC or PE).

NOTICE

Fittings within the HEC should be backed with a second wrench in order to prevent them from turning. Failure to do so could result in fluid leaks and damage to the equipment.

9. Using a 1-1/4" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting, remove the Port A hose from the Heat Exchanger (top, left fitting of heat exchanger when looking from the front). It may be necessary to cut the insulation around the fitting in order to expose the wrench flats. There may be residual coolant in the hose that will need to be drained.
10. Again using the 1-1/4" wrench and the adjustable wrench, remove the Port AB hose from Facility Return connection at the header. As above, insulation may need to be removed at the header and the hose drained.
11. Again using the 1-1/4" wrench and the adjustable wrench, remove the Port B hose from the Facility Supply connection at the header. As above, insulation may need to be removed at the header and the hose drained. To locate the correct hose, follow the hose connection from the valve. Refer to Illustration 5 for locations of Port B on the Facility Supply Header.

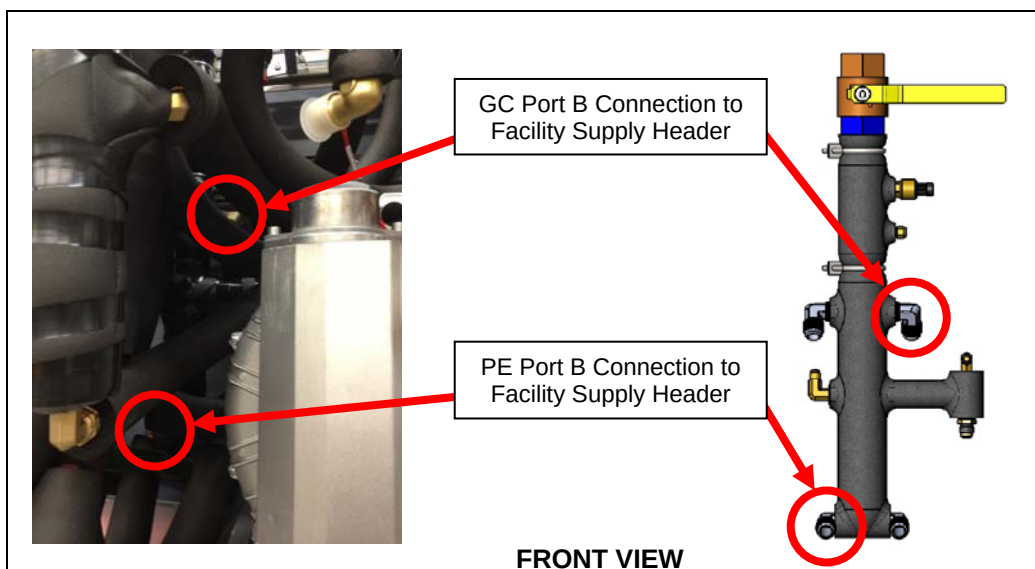


Illustration 5 – Front View Location of Port B Connections to Facility Supply Header for GC and PE Valves

12. **a. (G6000EN Cabinet Valve Replacement)** - While supporting the valve assembly with one hand, remove the 1/4-20 nuts that secure the valve to the system using a U bracket assembly using the 7/16" deep well socket or wrench. There are 2 U brackets, on each U bracket there are 2 nuts (Total of 4 nuts per valve). Refer to Illustration 6 for location of the 4 nuts for each of the valves.
- b. (G6001EN Cabinet Valve Replacement)** - Loosen hex screw on side of mounting bracket using the 7/16" deep well socket or wrench.

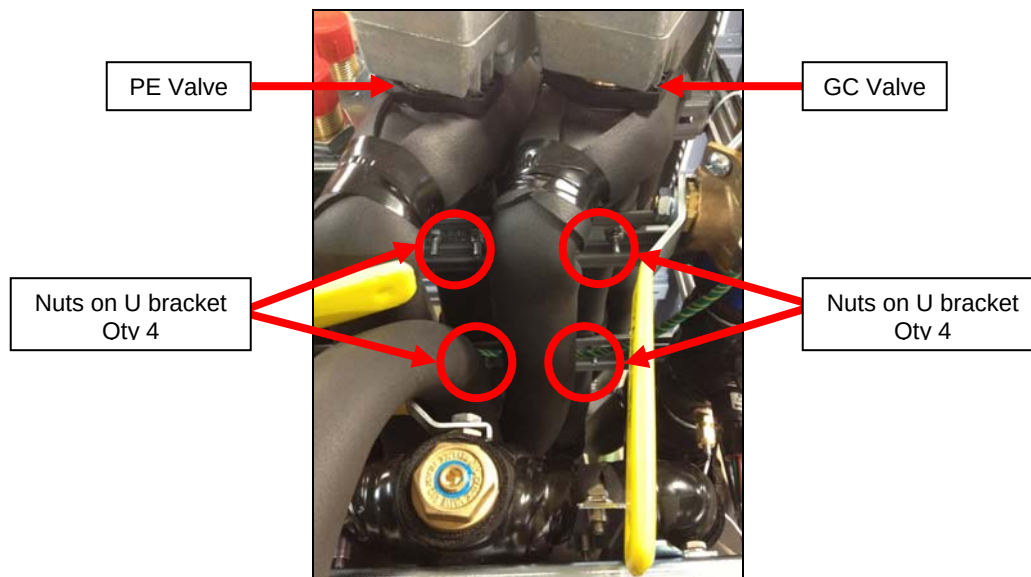


Illustration 6 – Location of 4 Nuts for Each of the GC and PE Valves

13. Remove the valve assembly (valve, 3 attached hoses, and bracket hardware) by pulling it up through the top of the cabinet. It may be necessary to alternate feeding the hoses up through the cabinet and pulling the valve assembly out the top of the cabinet.

Installation Procedure:

NOTICE

The mixing valve has a RED MANUAL/AUTO SWITCH. The valve must be placed in AUTO prior to energizing the system. Identify the correct position of the switch prior to

energizing system (Refer to Illustration 7). Verify prior to operating unit. Failure to place the switch in AUTO could result in failure of the unit to operate correctly.



Illustration 7 – Manual/Auto Switch Position on Mixing Valve

1. Insert the new valve assembly into the cabinet from the top by feeding the hoses down into the cabinet. Both style cabinets will have one hose (Port AB) inserted between the frame strut and cabinet side panel and the other two hoses (Ports A and B) will be inserted between the two frame struts on the left of the cabinet (Reference Illustration 8).



Illustration 8 – New Style Valve Installed in G6000EN Cabinet (System without Insulation)

2. Once in the cabinet, the four tabs of the mounting bracket should go behind the two frame struts. Secure the assembly to frame with hex screw on the left side of the bracket using the 7/16" deep well socket or wrench. (Reference Illustration 9)

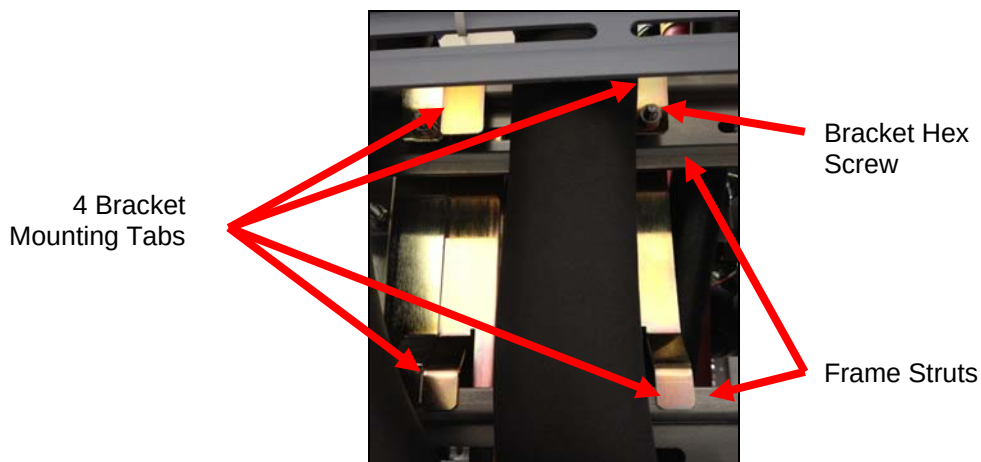


Illustration 9 – Rear View of PE Mixing Valve Installed in Cabinet

3. Using a 1-1/4" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting install the Port B hose to the Facility Supply connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.

4. Again using the 1-1/4" wrench and the adjustable wrench, install the Port A hose to the Heat Exchanger (top, left fitting of heat exchanger when looking from the front). Tighten hand tight and then tighten an additional 1/4 turn.
5. Again using the 1-1/4" wrench and the adjustable wrench, install the Port AB hose to Facility Return connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.
6. **(G6000EN Systems Only)** - After installation, the new mixing valves will require the fuse in the Signal Box to be changed to a higher value (T2AL250V). The fuse is part of the New Mixing Valve Kit and needs to be installed on the inside left hand side wall of the Signal Box. (Reference Illustration 10)

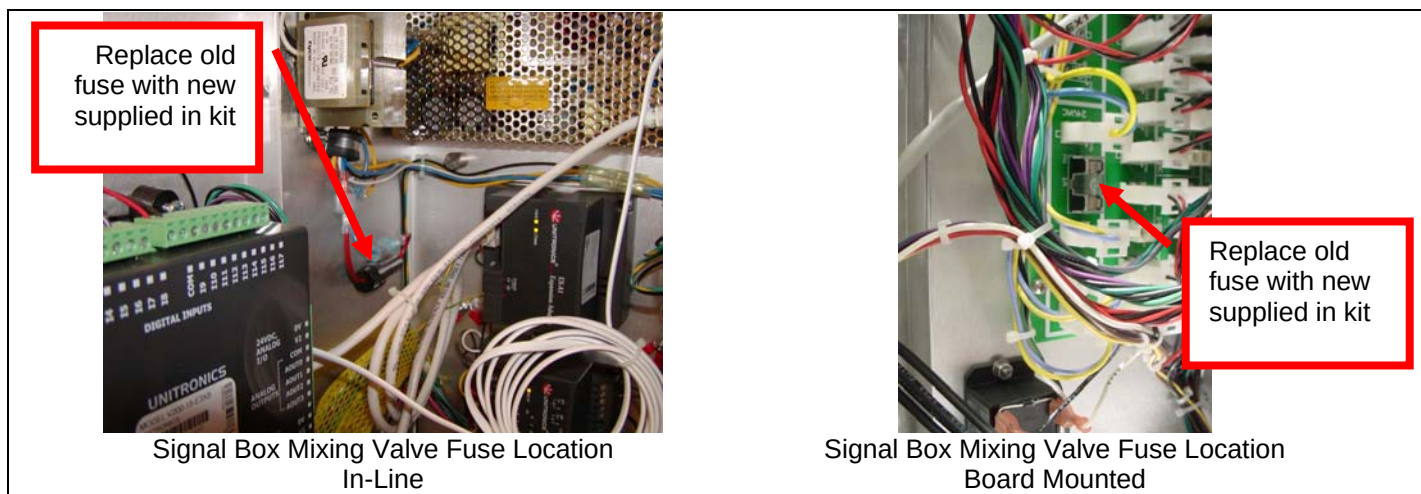


Illustration 10 – G6000EN Signal Box Fuse Locations (Two Design Styles)

7. Label the new fuse with the supplied label. For the inline fuse, place the label in the vicinity of the fuse. For the circuit board mounted fuse, place the label over the existing labeling on the circuit board.
8. Remove any tubing utilized for draining in previous steps.
9. Ensure all connections are tight and properly insulated.
10. Reconnect 5 pin connector to mixing valve assembly.
11. Open Facility Supply Ball valve.
12. Open Facility Return Ball valve.
13. Check for leaks around the manifolds of mixing valve(s) replaced.
14. Use the 2" insulation supplied to rewrap the terminations of the hose insulation within the system as necessary.
15. Reinstall any assemblies (Electrical Boxes, Doors, etc) that were removed during the process to gain access to mixing valves within the cabinet.
16. Clear Lockout/Tag out for the system in accordance with GE procedures.
17. Restart system.
18. Place system back into operation.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
1	Engineering Release	01/08/2013	GDD	
2	Changed HEC1 to G6000EN and HEC1.5 to G6001EN.	01/08/2013	GDD	
3	Updated with requested changes from GEHC. Changed required people to 2, made removal of the Signal Box or Electrical Box recommended in removal step 6, corrected valve callout in removal step 7, added illustrations to removal steps 11 and 12, and inserted installation step 7 to label the new fuse.	01/15/2013	GDD	
4	Revised and clarified dimensions in illustration 3.	01/18/2013	GDD	



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820-0340 – Replacement Instructions Chilled Air Blower Mixing Valves PN 205-0121-31 (GE P\N 5156320)

Purpose: The purpose of this document is to provide instruction on how to remove and replace the Chilled Air Blower mixing valve. Note that there are 2 different types of valves (refer to illustrations within this document to determine mixing valve type). The mixing valve will be supplied with a mounting bracket and hoses making installation and removal in all cabinets possible.

Scope: This instruction applies to all DVMR Heat Exchanger Cabinets (HEC).

Tools Required:

- 7/8" open end or adjustable wrench.
- Adjustable wrench (opening up to 3/4" for backing stationary fittings)
- 7/16" deep welled socket or wrench.
- Flat-head screwdriver (for removing cover of valve actuator for installation into G6000EN cabinets)
- Absorbent towels.
- Wet/Dry Vacuum or Bucket for draining.

People Required: 2

Time Required: 2 hours per valve.

Refer to the illustrations below to determine mixing valve type.

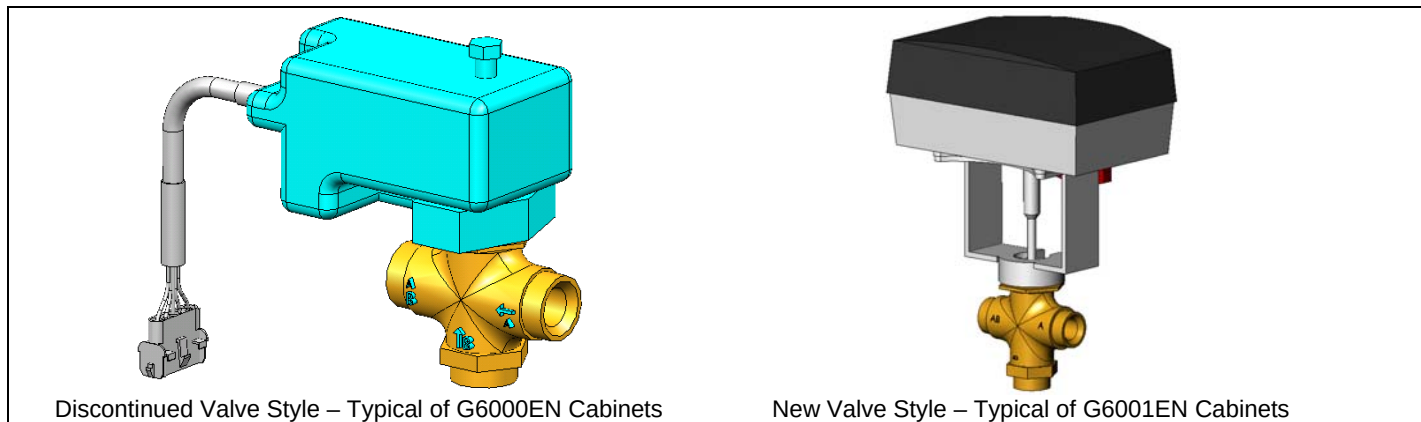


Illustration 1 – Mixing Valve Styles

Removal Procedure:

1. Perform shutdown and lockout of system in accordance with the GE "LOTO for Heat Exchanger Cabinet (HEC)" procedure.
2. Open door to heat exchanger cabinet. If necessary, the door can be removed from its hinges by disconnecting the ground strap (The ground strap nut can either be removed using a 10mm or adjustable wrench) and lifting up on the door.

3. Close Facility Supply and Return Ball valves located at the top of the HEC to isolate HEC from facility water.
4. Locate the chilled air mixing valve (Refer to Illustration 2).

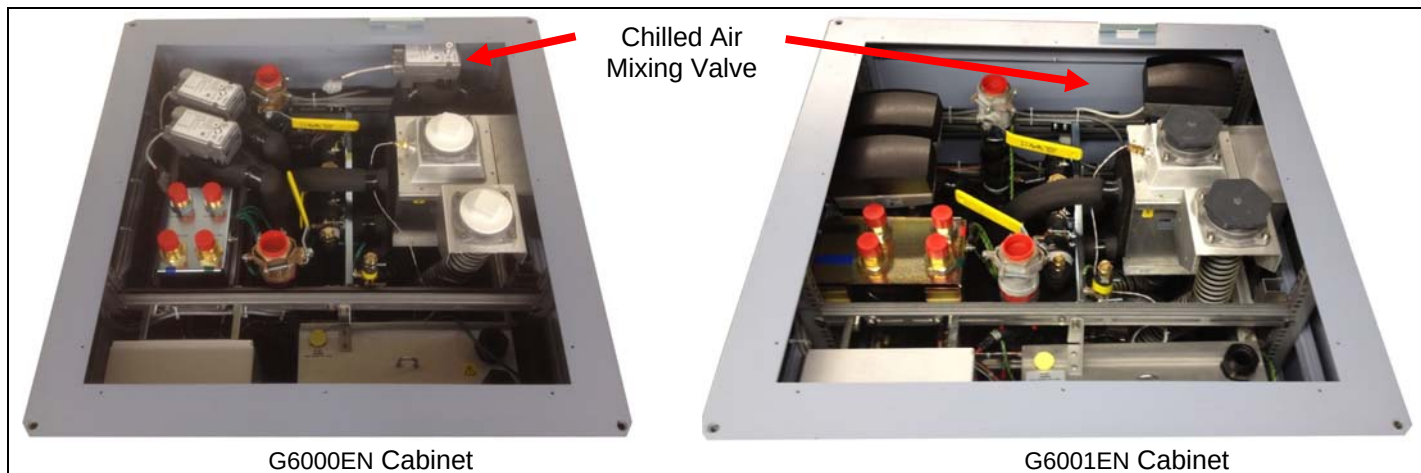


Illustration 2 – Chilled Air Mixing Valve Locations in G6000EN and G6001EN Cabinets

5. Drain Facility side plumbing per GE draining procedures.

Mixing Valve Assembly Kit

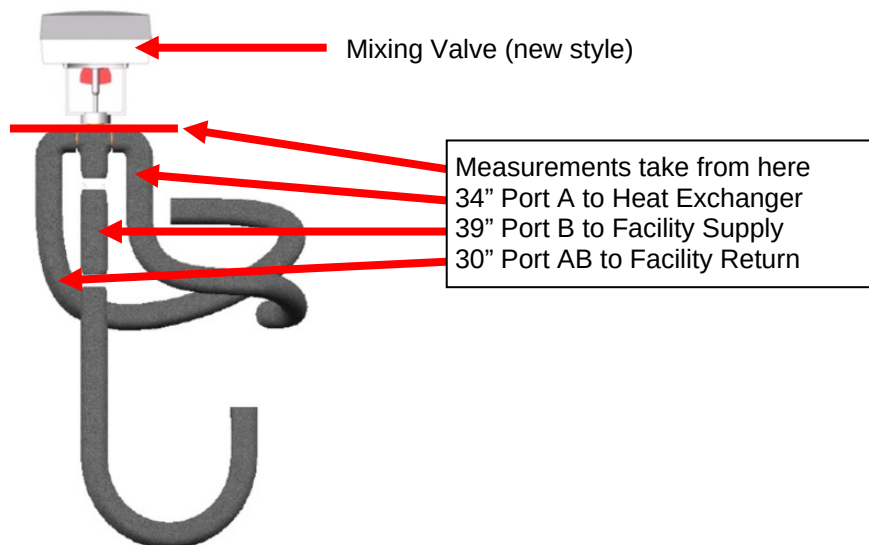


Illustration 3 - Connections for the Chilled Air Mixing Valve Kit



When removing items weighing greater than 35lbs, the hoist system must be utilized. Failure to use the hoist system may result in damage to the equipment or personal injury.

6. It is recommended to move/remove either the Signal Box or the Electrical Box to gain access to connection points on Facility Return header. In order to slide the Electrical Box to the side, disconnect the cables between the Electrical Box and the Signal Box. Refer to Lytron Procedures #820-0233 (Electrical Box) and #820-0234 (Signal Box) for details on removing these parts. It is also recommended to remove the blower to get better access to the hoses per GE main Blower replacement procedure under Heat Exchanger Cabinet Setup.

7. The Photos below detail the locations for connecting ports A, AB, and B to the system. Refer to Illustration 3 above for locations of ports detailed in following photos. NOTE: Both Signal Box and Electrical Box have been removed in the photos below for ease of locating connections.

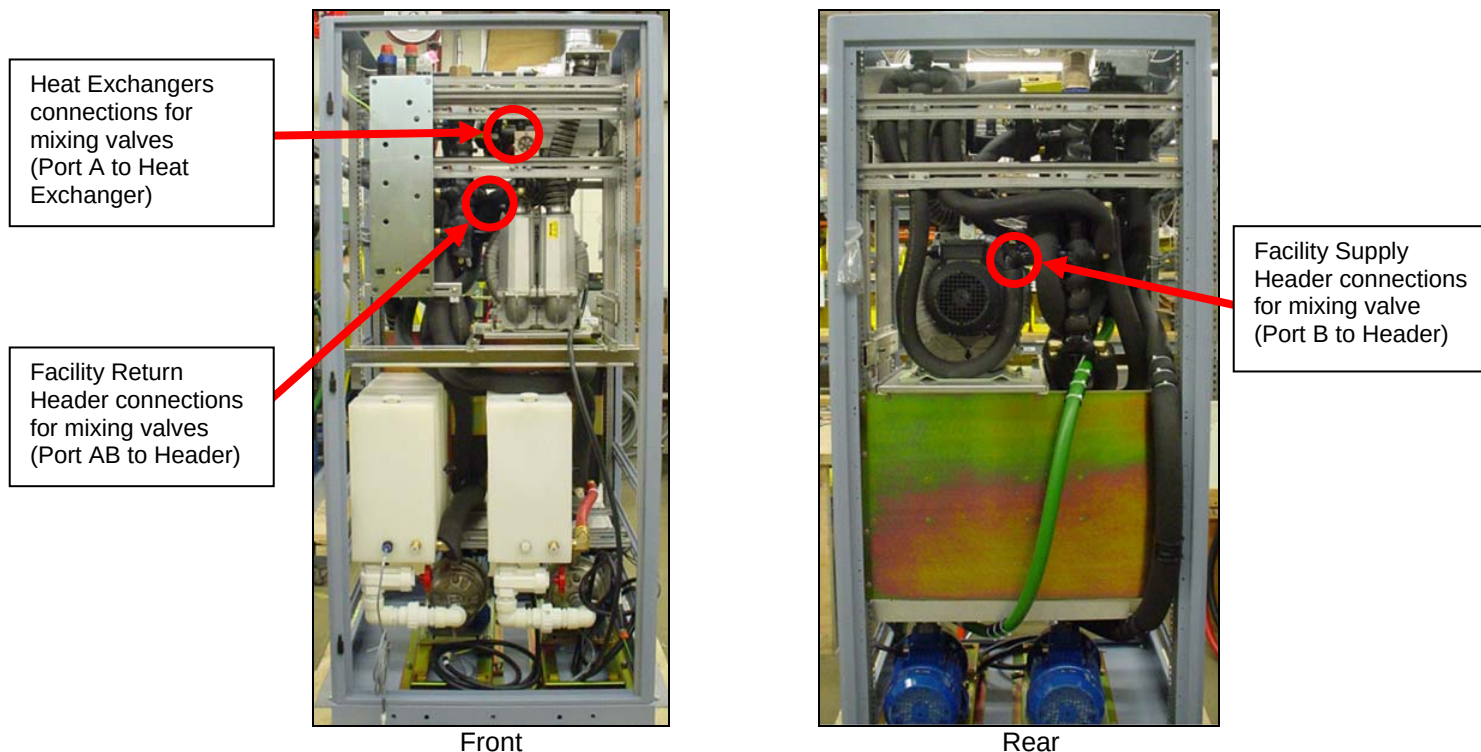


Illustration 4 - Detail of Cabinet with Signal Box and Electrical Box removed (Front and Rear Views)

8. Remove 5 pin connector for the mixing valve.

NOTICE

Fittings within the HEC should be backed with a second wrench in order to prevent them from turning. Failure to do so could result in fluid leaks and damage to the equipment.

9. Using a 7/8" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting, remove the Port A hose from the Heat Exchanger (top fitting of Liquid-to-Air heat exchanger). It may be necessary to cut the insulation around the fitting in order to expose the wrench flats. There may be residual coolant in the hose that will need to be drained (Reference Illustration 5)

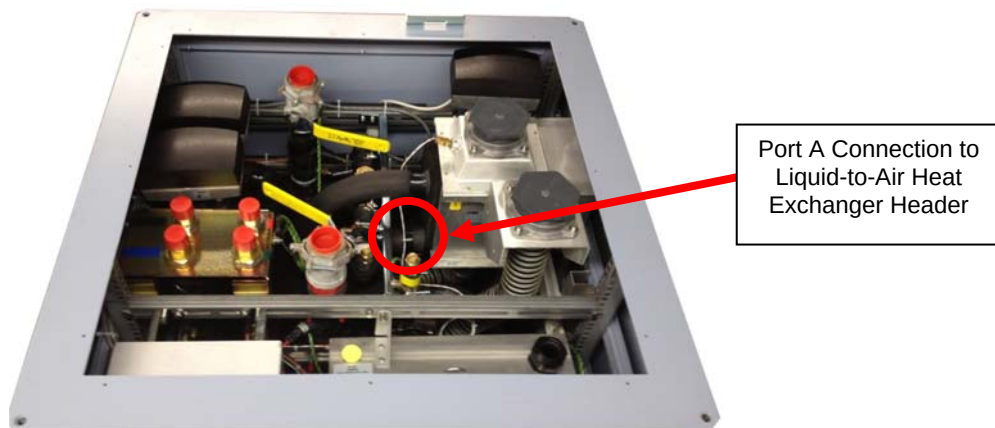


Illustration 5 - Port A Location of the Chilled Air Mixing Valve

10. Again using the 7/8" wrench and the adjustable wrench, remove the Port AB hose from Facility Return connection at the header. As above, insulation may need to be removed at the header and the hose drained.
11. Again using the 7/8" wrench and the adjustable wrench, remove the Port B hose from the Facility Supply connection at the header. As above, insulation may need to be removed at the header and the hose drained. To locate the correct hose, follow the hose connection from the valve. Refer to Illustration 6 for locations of Port B on the Facility Supply Header.

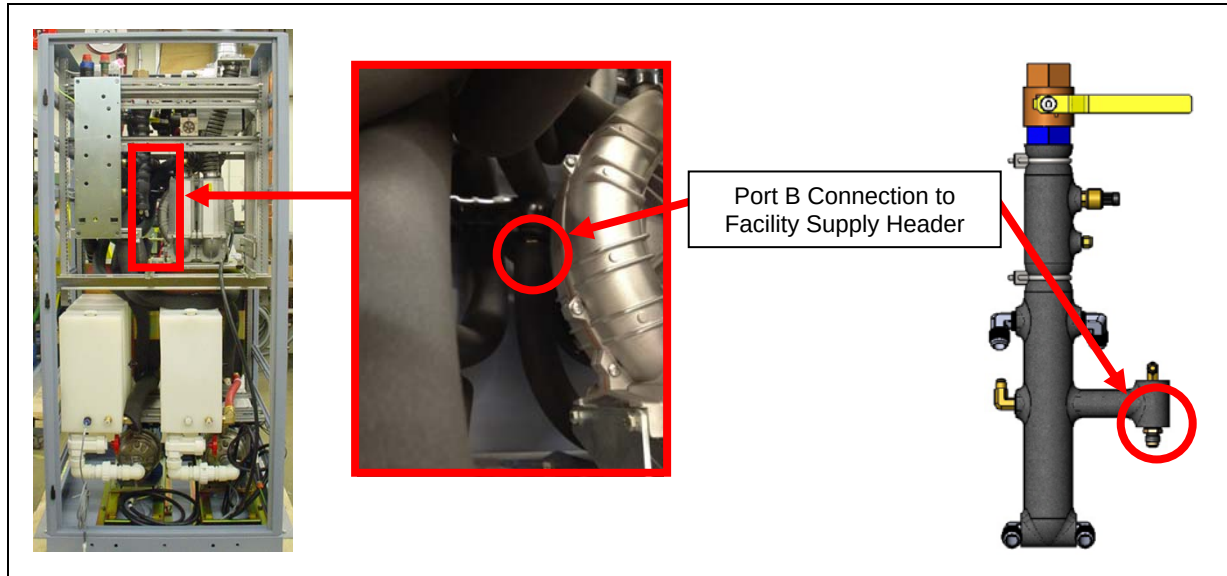


Illustration 6 – Front View Location of Port B Connection to Facility Supply Header

12. **a. (G6000EN Cabinet Valve Replacement)** - While supporting the valve assembly with one hand, remove the 1/4-20 nuts that secure the valve to the system using a U bracket assembly using the 7/16" deep well socket or wrench. There are 2 U brackets, on each U bracket there are 2 nuts (Total of 4 nuts per valve). Refer to Illustration 7 for location of the 4 nuts for each of the valves.
- b. (G6001EN Cabinet Valve Replacement)** - Loosen hex screw on side of mounting bracket using the 7/16" deep well socket or wrench.

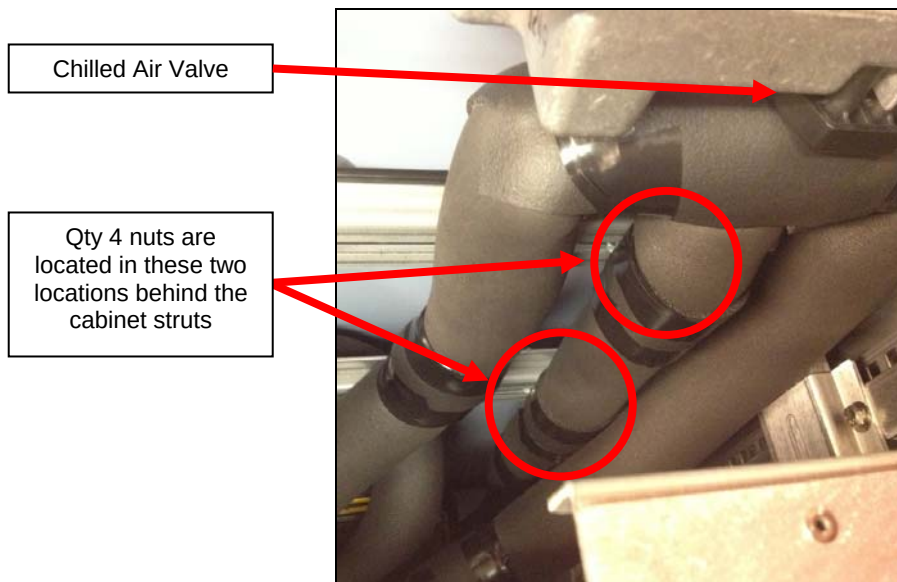


Illustration 7 – Location of 4 Nuts for Each of the GC and PE Valves

13. Remove the valve assembly (valve, 3 attached hoses, and bracket hardware) by pulling it up through the top of the cabinet. It may be necessary to alternate feeding the hoses up through the cabinet and pulling the valve assembly out the top of the cabinet.

Installation Procedure:

NOTICE The mixing valve has a RED MANUAL/AUTO SWITCH. The valve must be placed in AUTO prior to energizing the system. Identify the correct position of the switch prior to energizing system (Refer to Illustration 8). Verify prior to operating unit. Failure to place the switch in AUTO could result in failure of the unit to operate correctly.



Illustration 8 – Manual/Auto Switch Position on Mixing Valve

1. **(G6000EN Cabinet Valve Installation Only)** – It may be necessary to remove the top cover of the actuator that is part of the mixing valve assembly in order to fit the Chilled Air Valve into a G6000EN cabinet. In order to do this, insert a flat-head screwdriver under the tab of black actuator cover and lift up (Reference Illustration 9).

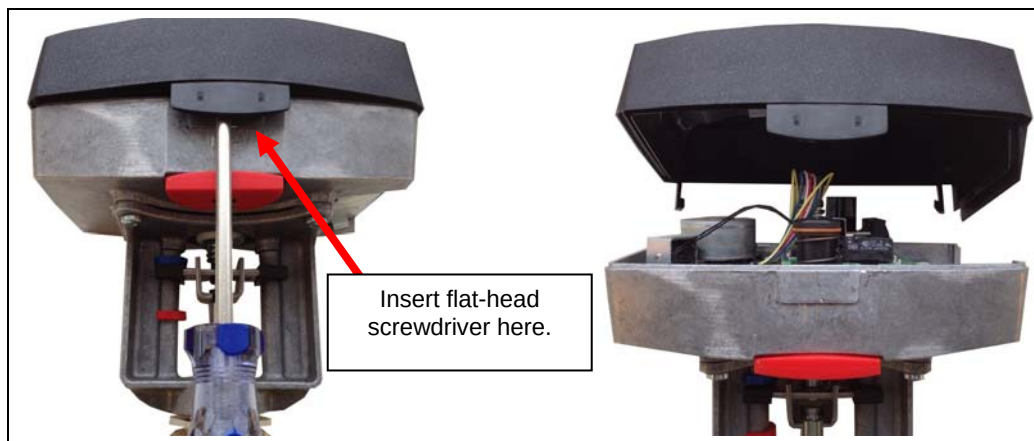


Illustration 9 – Removal of the Actuator Cover

2. Insert the new valve assembly into the cabinet from the top by feeding the hoses down into the cabinet. Both style cabinets will have the new valve installed such that all three hoses will be in front of the two cabinet struts that run horizontally across the rear of the cabinet.

3. Once in the cabinet, the four tabs of the mounting bracket should go behind the two frame struts. Secure the assembly to frame with hex screw on the left side of the bracket using the 7/16" deep well socket or wrench. (Reference Illustration 10)

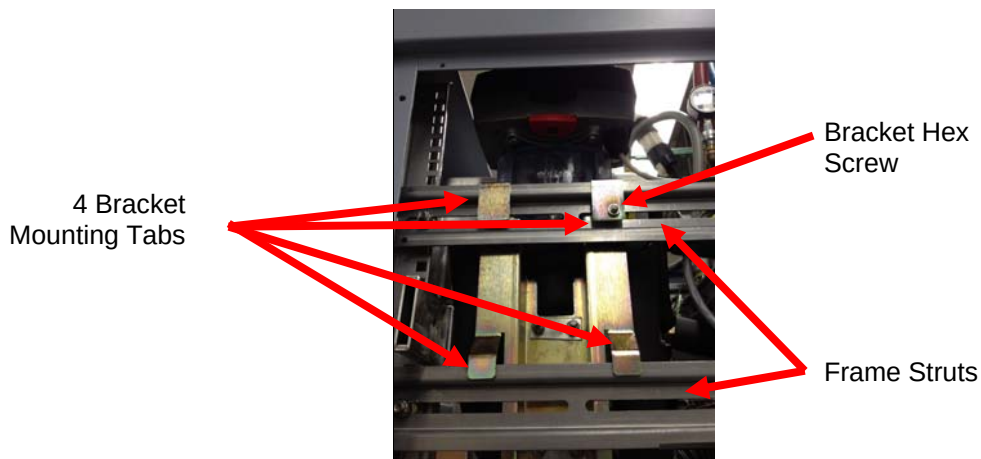


Illustration 10 – Rear View of Chilled Air Mixing Valve Installed in Cabinet

4. Once the valve assembly has been secured in place, re-install the actuator cover if removed during step 1.
5. Using a 7/8" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting install the Port B hose to the Facility Supply connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.
6. Again using the 7/8" wrench and the adjustable wrench, install the Port A hose to the Heat Exchanger (top fitting of Liquid-to-Air heat exchanger). Tighten hand tight and then tighten an additional 1/4 turn.
7. Again using the 7/8" wrench and the adjustable wrench, install the Port AB hose to Facility Return connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.
8. **(G6000EN Systems Only)** - After installation, the new mixing valves will require the fuse in the Signal Box to be changed to a higher value (T2AL250V). The fuse is part of the New Mixing Valve Kit and needs to be installed on the inside left hand side wall of the Signal Box. (Reference Illustration 11)

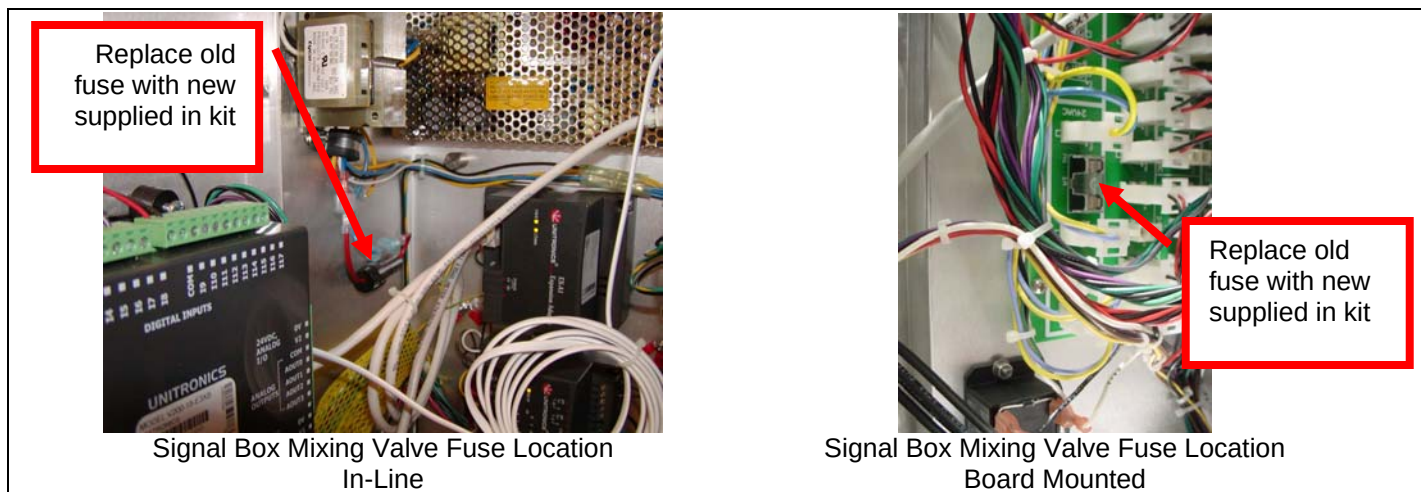


Illustration 11 – G6000EN Signal Box Fuse Locations (Two Design Styles)

9. Label the new fuse with the supplied label. For the inline fuse, place the label in the vicinity of the fuse. For the circuit board mounted fuse, place the label over the existing labeling on the circuit board.

10. Remove any tubing utilized for draining in previous steps.
11. Ensure all connections are tight and properly insulated.
12. Reconnect 5 pin connector to mixing valve assembly.
13. Open Facility Supply Ball valve.
14. Open Facility Return Ball valve.
15. Check for leaks around the manifolds of mixing valve(s) replaced.
16. Use the 2" insulation supplied to rewrap the terminations of the hose insulation within the system as necessary.
17. Reinstall any assemblies (Electrical Boxes, Doors, etc) that were removed during the process to gain access to mixing valves within the cabinet.
18. Clear Lockout/Tag out for the system in accordance with GE procedures.
19. Restart system.
20. Place system back into operation.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
1	Engineering Release	01/16/2013	GDD	
2	Changed Illustration 10 to say Chilled Air and updated arrows in illustration 3.	01/17/2013	GDD	
3	Revised and clarified dimensions in illustration 3.	01/18/2013	GDD	
4	Corrected location of Port A in illustrations 4 and 5.	01/30/2013	GDD	



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820-0234 Replacement Instructions - Signal Box (PN 205-0121-06) GE PN 5269691

Tools Required:

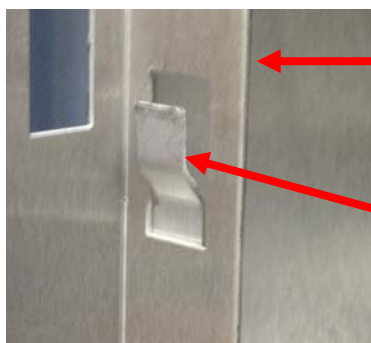
- Flat bladed screwdriver
- Wire Cutters(as needed to cut zip ties)

NOTICE

The signal box weighs approximately 14 lbs.

Removal Procedure:

1. Perform shutdown and lockout of system in accordance with the GE 'LOTO for Heat Exchanger Cabinet (HEC) procedure.
2. Open door to heat exchanger cabinet. If necessary, the door can be removed from its hinges by lifting up on the door.
3. Open front cover of signal box by loosening 6 thumbscrews.
4. Secure signal box cover to Signal box frame by hanging cover hinge holes over the Signal frame built-in tabs. Refer to Photo 1.



Signal box cover with hinge holes.

Signal box frame with built-in tab.

Photo 1 Detail showing mounting Signal Box cover onto built-in tabs of frame.

5. Remove all cables from their respective J## sockets -on the right side of the signal box.

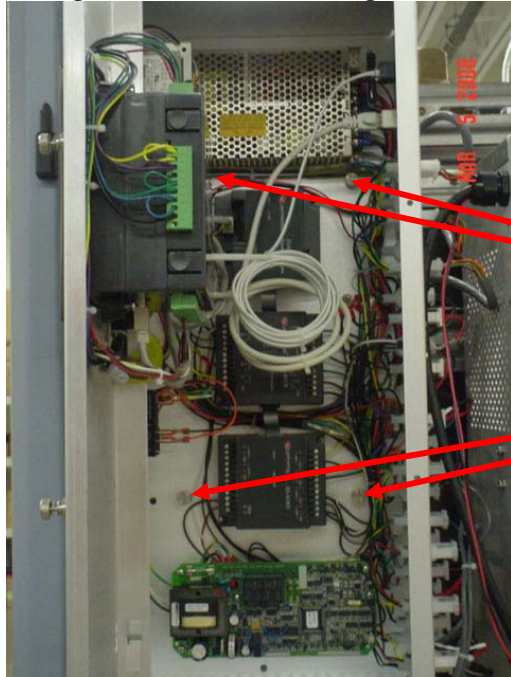


Photo 2 Detail of different connection locations



Photo 3 Detail showing HEC ID connector

6. While supporting the signal box with one hand, loosen 4 thumbscrews located inside the signal box that secure the signal box to the heat exchanger cabinet frame.



Thumbscrews – 1 in
each corner

Photo 4 signal box interior- thumbscrew location

7. Close the main cover to the signal box.
8. Crate signal box for shipment.

End of Procedure

Installation Procedure:

1. Remove new signal box front cover by loosening 6 screws.
2. Position the signal box so that the mounting holes in the main cabinet frame align with thumbscrews on the signal box.
3. Tighten one thumbscrew at a time to just finger tight. Tighten thumbscrew and then its opposite counterpart until all 4 thumbscrews are secure. Refer to attached photo 4.
4. Plug connectors J1 through J28 into the appropriate jack on the right side of the signal box. Refer to attached photo 1.
 - o Depending on revision of signal box, the location of the connectors J1 through J28 may be different (see attached photo 2). Later revision signal box will have a J29 'HEC ID' plug which will only have a connector plug on G6001EN cabinets (see attached photo 3). J10 'DI-2' and J27 'BLW-VEL' may also remain blank.
 - i. NOTE: When replacing the new signal box in a cabinet that had an earlier revision signal box some cable ties may have to be cut. The power box door will have to be opened to cut the J1 zip ties to gain enough length on the cable to make the connection (see attached photo 5).



Photo 5 Close-Up of J1 zip ties in power box

5. Replace front cover on the signal box.
6. Remove Lockout/Tagout in accordance with GE procedures.
7. Start system per Lytron manual #820-0190.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
C		1/24/2013		